What is range() in Python?

The **range()** function is used to generate a **sequence of numbers**.
It's commonly used in **loops** — like for loops — to repeat code a specific number of times.

🔹 **Syntax:**

range(start, stop, step)

Parameter	Description	Default
start	The number to start from	0
stop	The number to stop **before** (not included)	—
step	The difference between each number	1

🔸 **1. Using One Argument – range(stop)**

Starts from **0** and ends at **stop - 1**

```
for i in range(5):
    print(i)
```

🟢 Output:

0

1

2

3

4

🔸 **2. Using Two Arguments – range(start, stop)**

Starts from **start**, ends **before stop**

```
for i in range(2, 7):
```

```
    print(i)
```

🟢 Output:

```
2
3
4
5
6
```

🔸 **3. Using Three Arguments – range(start, stop, step)**

The third argument defines the **step (increment/decrement)** value.

```
for i in range(1, 10, 2):
    print(i)
```

🟢 Output:

```
1
3
5
7
9
```

🔸 **4. Negative Step – Counting Backward**

You can use a **negative step** to count backward.

```
for i in range(10, 0, -2):
    print(i)
```

🟢 Output:

```
10
8
```

6

4

2

🔸 **5. Using range() in a List**

You can convert a range into a list or tuple.

```
nums = list(range(5))

print(nums)
```

🟢 Output:

[0, 1, 2, 3, 4]

```
nums = tuple(range(2, 10, 3))

print(nums)
```

🟢 Output:

(2, 5, 8)

🔸 **6. Example: Using range() in a Loop**

```
for i in range(1, 6):
    print("Python", i)
```

🟢 Output:

Python 1

Python 2

Python 3

Python 4

Python 5

🧱 7. Example: Sum of Numbers using range()

```
total = 0
for i in range(1, 6):
    total += i
print("Sum:", total)
```

🟢 Output:

```
Sum: 15
```

📜 Summary Table

Form	Description	Example	Output
range(stop)	0 → stop-1	range(5)	0,1,2,3,4
range(start, stop)	start → stop-1	range(2, 6)	2,3,4,5
range(start, stop, step)	step between values	range(1, 10, 2)	1,3,5,7,9
Negative step	Counts backward	range(10, 0, -2)	10,8,6,4,2